

BUSINESS USE CASE

Firestore Data Connect

Powering a Modern E-Commerce Platform

From Schema to SDK — Real-time, Type-safe, Cross-platform

Presented to Engineering & Leadership

What Is Firebase Data Connect?



SQL-Powered Backend

A fully managed PostgreSQL database hosted on Google Cloud — no servers to provision or maintain.



GraphQL API Layer

Define your data shape once in GraphQL. Firebase auto-generates all the query and mutation logic.



Auto-Generated SDKs

One schema generates type-safe client SDKs for Web, Flutter, iOS & Android — no manual API code.

One unified platform — from database to device, fully managed by Google.

Our Use Case: ShopNow — E-Commerce Platform

What We're Building

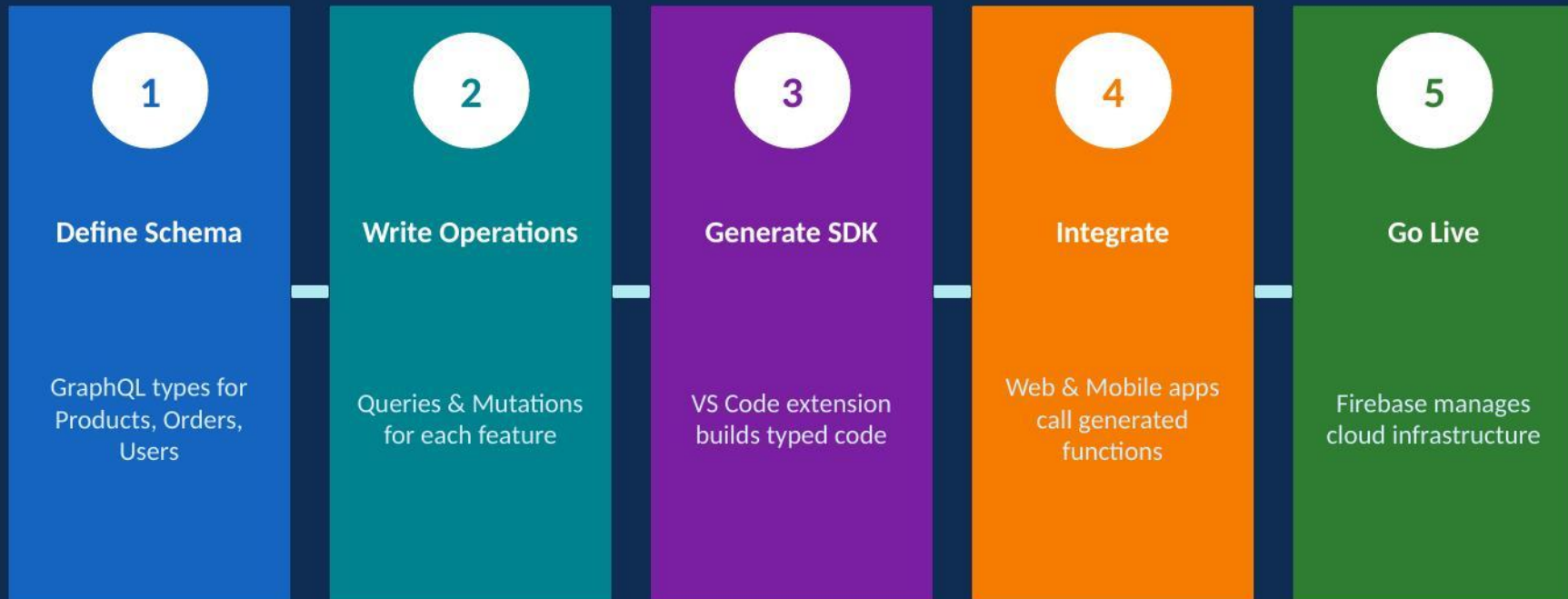
- Product catalog with search & filters
- User accounts & wishlist
- Shopping cart & checkout flow
- Order tracking & history
- Admin dashboard for inventory

Why Firebase Data Connect?

- Single schema for web + mobile apps
- No backend API team needed
- Real-time updates out of the box
- Scales automatically with traffic
- Secure by default with Firebase Auth

How It All Fits Together

A simple flow from schema definition to live app



Local emulator mirrors production — developers test everything before deploying to Google Cloud.

Delivery Plan — Development Phases

Phase 1

Environment Setup

- Install Node.js & Firebase CLI
- VS Code Firebase Data Connect extension
- Docker Desktop + PGlite for local DB
- Boot Firebase Emulator locally

Phase 2

Schema & Data Design

- Draft GraphQL schema (Products, Orders, Users, Cart)
- Define Queries (list, get, search)
- Define Mutations (create, update, delete)
- Test all operations in VS Code

Phase 3

Web App Integration

- Set SDK output path in dataconnect.yaml
- Run npm install firebase
- Call generated TypeScript functions in React

Phase 4

Mobile App Integration

- Add Firebase Data Connect plugin (pubspec.yaml / build.gradle)
- Generate platform SDK via VS Code
- Point mobile app to emulator on port 9399